

CMCP-Guided ePC Distillation

Matched-Budget Calibration: Setup, Results, and Weighting Diagnostics

RelaLeap experiment record 20260725T161000Z

25 July 2026

Executive summary

This was a small, local, three-seed calibration of typed CMCP evidence accounting inside a predictive-coding (ePC) distillation objective. It was designed to answer a narrow question: can CMCP prevent a burst of duplicate/derived teacher messages from being counted as four independent pieces of evidence, while retaining a deliberately independent teacher view?

Observed. CMCP did the duplicate-protection part cleanly: its mean effective precision was 1.354 instead of naive accumulation’s 4.0. It also improved final Task-B loss relative to naive accumulation in all three paired seeds (mean difference -0.0190 nats). However, CMCP gave the constructed independent view only about 0.354 additional credit, whereas the constructed oracle credited it fully and therefore had total precision 2.0. CMCP was worse than ordinary one-view KD/ePC on Task-B loss in all three seeds (mean $+0.00341$ nats), while its Task-A retention was only slightly better (mean -0.00144 nats).

Conclusion. The run supports CMCP as a guard against duplicate precision inflation. It does not support a claim that the present response score or raw weighting improves transformer-to-ePC distillation. The most important issue is a confound: CMCP changes both *which* packet is trusted and the *total KD objective mass*. The next test should hold the total non-duplicate KD mass fixed while comparing novelty estimators.

Question and scope

Under equal optimizer-update and teacher-call budgets, does typed CMCP accounting prevent duplicate/derived teacher packets from inflating credit, retain an independent view, and improve calibration or sequential-curriculum retention for an ePC student?

This is a synthetic, reduced calibration—not a transformer-scale result and not a confirmation experiment. It uses a two-layer causal character transformer with vocabulary 12, width 16, two attention heads, feed-forward width 32, sequence length 16, and dropout 0. The teacher and student are different models with a shared fixed architecture. The teacher is trained for 32 AdamW updates on the forward synthetic sequence task and then frozen. Each student starts from the same copied initialization within a seed.

Exact protocol

Curriculum and budgets

Each seed runs 160 AdamW student updates: 80 Task-A updates followed by 80 Task-B updates. A batch contains 6 sequences; evaluations use 6 batches of 8 sequences. Task-A is a stochastic forward modular-increment sequence task. Task-B uses another generated stream but its targets are shifted by half the vocabulary: $y_B = (y + 6) \bmod 12$. Thus Task B contradicts the Task-A target mapping rather than merely continuing it.

The seeds are 101, 211, and 307. Every arm receives exactly 160 optimizer updates, 160 teacher forward calls, and 640 packet presentations (four per update). The ePC distillation objective uses two settlement steps, temperature 2.0, relaxation learning rate 0.1, and the overall objective

$$L = 0.8L_{CE}(y, f_{\theta}(x)) + 0.02 \sum_p w_p L_{ePC}(f_{\theta}(x), z_p).$$

Gradients are clipped at norm 1.0. All recorded ePC energy traces were monotone in every arm and seed.

Teacher packet burst

For each student batch, the frozen teacher emits base logits z . Four typed packets, all for the same cohort/batch, are presented:

Packet	Payload	Typed innovation/provenance relation	Constructed oracle credit
Base	z	canonical innovation; teacher/base	1
Exact duplicate	z copied	same innovation as base; teacher/cache-copy	0
Descendant	$z/2$	same innovation as base; teacher/temp-transform	0
Seeded view	$z + 0.35\epsilon$	distinct innovation; teacher/seeded-view	1

Here ϵ is a deterministic seed-and-update-specific Gaussian tensor. The oracle designation is a fixture assumption: base and seeded view are counted as two independent pieces of evidence even though the latter is an additive perturbation of the former. That assumption is useful for diagnosing under-credit, but it is not a claim that noisy logits are generally independent teacher knowledge.

Arms and exact weights

Arm	Packet weights in burst order	Total effective precision	Purpose
Ordinary	(1, 0, 0, 0)	1	one canonical teacher target
Naive	(1, 1, 1, 1)	4	unprotected multi-view accumulation
Oracle	(1, 0, 0, 1)	2	fixture-known deduplication target
CMCP	(1, 0, 0, r)	$1 + r$	typed guards plus estimated novelty

CMCP first rejects later packets with an already seen innovation identity. For a new innovation, it converts logits to a flattened standardized response score: softmax, flatten batch and position, subtract each vocabulary-column mean, divide by its population standard deviation (floor 10^{-5}). It then uses the existing ridge-regularized conditional-information-ratio routine against stored canonical scores, clamps the ratio to $[0, 1]$, and assigns that ratio as the packet weight. Hence base has weight 1 and the seeded view has weight r ; duplicate and descendant have weight 0 by typed identity.

Measurements

Task-A retention is the final Task-A loss minus the Task-A loss immediately after update 80; lower is better. Task-B metrics are evaluated after update 160. Expected calibration error (ECE) uses ten fixed bins $[0, 0.1), \dots, [0.9, 1.0)$. Effective precision is the update-average sum of packet weights.

Aggregate results (mean across three paired seeds)

Arm	Precision	A retention Δ loss	B loss	B accuracy	B ECE
Ordinary	1.0000	0.16394	2.30898	0.14887	0.01092
Naive	4.0000	0.15676	2.33139	0.14931	0.00962
CMCP	1.35412	0.16250	2.31240	0.15148	0.00934
Oracle	2.0000	0.16003	2.31836	0.14714	0.00468

The CMCP seeded-view novelty weights were 0.32867, 0.39223, and 0.34146 for seeds 101, 211, and 307 respectively (mean 0.35412). Thus it eliminated the two typed redundancies, but recovered only 35.4% of the fixture’s additional independent-view credit.

Per-seed paired outcomes

Seed	CMCP precision	$L_B^{CMCP} - L_B^{naive}$	$L_B^{CMCP} - L_B^{ordinary}$	$\Delta A^{CMCP} - \Delta A^{naive}$	$\Delta A^{CMCP} - \Delta A^{ordinary}$	CMCP B acc.	CMCP B ECE
101	1.32867	-0.01771	+0.00332	+0.01904	-0.00357	.14714	.00633
211	1.39223	-0.01561	+0.00284	+0.00002	-0.00066	.15104	.01264
307	1.34146	-0.02366	+0.00408	-0.00185	-0.00010	.15625	.00904
Mean	1.35412	-0.01899	+0.00341	+0.00574	-0.00144	.15148	.00934

Negative loss differences favor CMCP. CMCP beats naive Task-B loss in all three seeds, but loses to ordinary Task-B loss in all three. Its retention is slightly better than ordinary in all three, but the magnitude is tiny on this toy scale. With only three calibration seeds and no frozen confirmation threshold, these are directions for diagnosis, not confidence-qualified efficacy claims.

Interpretation and alternative explanations

What is directly supported. Typed innovation identities prevent both the byte-identical packet and the declared descendant from being counted again. This reduces the pathological four-fold KD accumulation. CMCP’s lower Task-B loss versus naive is consistent with avoiding that overcount.

What is not supported. Neither the response-score novelty estimator nor raw CMCP weighting demonstrated a benefit over ordinary one-view KD/ePC. The oracle did not dominate ordinary either, so the fixture is not yet a validated setting in which “more genuinely independent teacher evidence” should improve this student outcome.

Primary confound. Raw CMCP has total KD weight about 1.354, while the oracle has 2 and naive has 4. It therefore jointly changes novelty selection and global distillation strength. Its result can arise from less KD, from better selection, or their interaction. The CE term is unchanged, but the relative CE/KD balance is not.

Why the independent view was underweighted. The view is defined as $z + 0.35\epsilon$, so its score remains strongly conditionally predictable from z . The current estimator measures residual response variation, not the fixture’s declared causal independence. This may be exactly the desired answer for a noisy re-expression of one teacher, but disagrees with this fixture’s oracle label. It should be resolved by a testable definition of independent teacher information, not by simply increasing the weight.

Other limitations. The descendant $z/2$ is declared redundant by identity without testing whether temperature-scaled logits have a distinct optimization effect. The same frozen teacher generates base and seeded view; there is no independent teacher checkpoint, data shard, augmentation pipeline, or causal intervention. Task B’s supervised labels contradict the forward teacher’s task, so it is deliberately a stress test of plasticity, not a natural multi-teacher distillation problem. Accuracy is close to the 1/12 chance regime; loss and ECE are more informative here, but also small and variable.

Weighting improvements worth testing

1. **Factor identity, novelty, and total mass.** Keep hard typed deduplication as a zero/one guard. Separately estimate novelty only for distinct innovations, then normalize the accepted weights to a predeclared total KD

mass (for example 2 in this fixture). Compare raw CMCP, normalized CMCP, and an equal-mass ordinary control. This makes the causal question “which evidence” rather than “how much KD.”

2. **Estimate incremental *training signal*, not raw-logit residual.** For each candidate packet, residualize its distillation gradient or ePC parameter-credit vector against the accepted packets and weight its positive residual norm. A temperature transform or noisy logit view can be redundant in probability space but non-redundant in parameter-update space, which is the quantity relevant to assimilation.
3. **Calibrate novelty on a fixture family with known fractions.** Vary the independent-view mixture continuously, e.g. $z' = \sqrt{1 - \rho^2}z + \rho u$ for a separately trained teacher u , and test whether estimated weight tracks the known incremental source fraction. Include exact copies, temperature transforms, paraphrase/augmentation views, independent checkpoints, and adversarially correlated views. Select the estimator before running a student outcome test.
4. **Treat descendant semantics explicitly.** Keep a provenance rule for deterministic cache copies, but make temperature, truncation, and augmentation descendants eligible for a conditional incremental test. Compare “always zero descendants” with “gradient-novel descendants” under equal total KD mass.
5. **Use a gated assimilation/maintenance split.** A packet with low novelty need not be discarded: send it to replay/maintenance credit while using only a novelty-normalized fraction for current parameter assimilation. Evaluate both separately; otherwise repeated packets can look useless even if they stabilize a learned representation.

Recommended next experiment

Before any large transformer run, use a 5–10 seed synthetic calibration with frozen primary metrics: (i) calibration error of estimated incremental credit against a known mixture fraction; (ii) paired final Task-B loss; (iii) Task-A retention; and (iv) energy monotonicity. Hold optimizer updates, teacher calls, and total KD mass constant. Compare ordinary-one-view, naive, oracle, raw-CMCP, mass-normalized CMCP-response, and mass-normalized CMCP-gradient-novelty. Promote to a real teacher/student only if the estimator first tracks known incremental credit and then improves a preregistered paired student metric over equal-mass ordinary KD/ePC.

Reproducibility and provenance

Implementation commit: `1afb6e1` on branch `agent/cmcp-epc-kd-bridge`. Principal implementation: `src/relaleap/hdpc/cmcp_distillation.py`. Runner: `scripts/run_cmcp_epc_kd_bridge.py`. Focused tests: `tests/test_cmcp_distillation.py`. Recorded command: `command.sh`. Final calibration JSON artifacts: `artifacts/calibration-seed{101, 211, 307}-u160-contradictory-v4.json`.

The focused compatibility suite recorded for the implementation was 37 passed; the full repository suite subsequently passed 416 tests with 1 skip in the project virtual environment. This report is derived directly from the three final v4 JSON artifacts and the experiment RUN record; earlier 40-update revisions remain retained but are not pooled into the results above.